BASE CHANGE COMPATIBILITY OF THE EXCHANGE MAP BETWEEN LOWER SHRIEK ALONG SEPARATED LOCALLY QUASI-FINITE MORPHISMS AND RF_* ALONG PROPER MORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Disclosure: two steps are carried by displayed data and standard K-injective formalism rather than a dedicated argument. (1) The homotopy-commutative square of quasi-isomorphisms (proof lines 3107-3116) is picked by the author with the equality arrow sourced to Lemma lemma-proper-f-shriek; the existence of the homotopy is the standard K-injective argument and is not spelled out. (2) The reduction from the shriek functors to the star functors (proof lines 3141-3146) rests on the shriek functors being subfunctors of the star functors with base change maps defined compatibly, anchored by a pointer to the proof of Lemma lemma-base-change-f-shriek-separated. The final commutativity of the last displayed square is attributed to the choice of maps made earlier (proof lines 3174-3175). No verbatim 'we omit' occurs in the proof.

Allowed standard prerequisites (editorial).

- Exchange map for proper f and separated locally quasi-finite g (Lemma lemma-shriek-proper-and-open, canonical tag 0F7B). cited at statement line 3081 and proof lines 3094-3102. Supplies the vertical arrows of the 2x3 diagram; its proof contains the 4-term construction of the exchange map whose base change compatibility is being verified.
- Base change for separated locally quasi-finite lower shriek (Lemma lemma-base-change-f-shriek-separated, canonical tag 0F55). cited at statement line 3085 and proof lines 3141-3146. Source of the base change map for g in the horizontal arrows, and the anchor for the reduction from shriek to star functors.
- Sites stacking remarks (Remarks sites-remark-compose-base-change and sites-remark-compose-base-change-horizontal, canonical tags 0F6X and 0F6Y). cited at proof lines 3147-3150. The engine for the commutativity: pullback maps are compatible with horizontal and vertical stacking of base change diagrams.

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Source reference. Original reference 0EZQ: sources/stacks-chow/tags.txt, source lines 18735–18735. Cited at statement lines 3082-3084: source of the base change map for the proper morphism f appearing in the horizontal arrows.

Source reference. Original lemma 0F55: sources/stacks-chow/tags.txt, source lines 18925–18925. Cited at statement line 3085 and proof lines 3141-3146: source of the base change map for g, and the anchor for the shriek-to-star reduction.

Source reference. Original lemma 0F51: sources/stacks-chow/tags.txt, source lines 18921–18921. Cited at proof line 3118: justifies the equality arrow g'_{1,!}(a')^{-1} = a^{-1}g'_! in the homotopy-commutative square.

Source reference. Original lemma 0F7B: sources/stacks-chow/tags.txt, source lines 19001–19001. Cited at statement line 3081 and proof lines 3094-3102: supplies the vertical arrows and the 4-term construction of the exchange map.

Source reference. Original reference 0F6X: sources/stacks-chow/tags.txt, source lines 18987–18987. Cited at proof lines 3147-3150: compatibility of pullback maps with vertical stacking, used for the star-functor commutativity.

Source reference. Original reference 0F6Y: sources/stacks-chow/tags.txt, source lines 18988–18988. Cited at proof lines 3147-3150: compatibility of pullback maps with horizontal stacking, completing the same argument.

Lemma 0.2. *Let $b : Y_1 \to Y$ be a morphism of schemes. Consider a commutative diagram of schemes*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array} \quad \text{and let} \quad \begin{array}{ccc} X'_1 & \xrightarrow{g'_1} & X_1 \\ {\scriptstyle f'_1}\downarrow & & \downarrow{\scriptstyle f_1} \\ Y'_1 & \xrightarrow{g_1} & Y_1 \end{array}$$

be the base change by b. Assume f and f' proper and g and g' separated and locally quasi-finite. For a ring Λ and K in $D(X'_{étale}, \Lambda)$ there is commutative diagram

$$\begin{array}{ccccc} b^{-1}g_!Rf'_*K & \longrightarrow & g_{1,!}(b')^{-1}Rf'_*K & \longrightarrow & g_{1,!}Rf'_{1,*}(a')^{-1}K \\ \downarrow & & & & \downarrow \\ b^{-1}Rf_*g'_!K & \longrightarrow & Rf_{1,*}a^{-1}g'_!K & \longrightarrow & Rf_{1,*}g'_{1,!}(a')^{-1}K \end{array}$$

in $D(Y_{1,étale}, \Lambda)$ where $a : X_1 \to X$, $a' : X'_1 \to X'$, $b' : Y'_1 \to Y'$ are the projections, the vertical maps are the arrows of Lemma 0F7B and the horizontal arrows are the base change map (from Étale Cohomology, Section 0EZQ) and the base change map of Lemma 0F55.

Proof. Represent K by a K-injective complex $\mathcal{J}^\bullet$ of sheaves of Λ-modules on $X'_{étale}$. Choose a quasi-isomorphism $g'_!\mathcal{J}^\bullet \to \mathcal{I}^\bullet$ to a K-injective complex $\mathcal{I}^\bullet$ of sheaves of Λ-modules on $X_{étale}$. The proof of Lemma 0F7B constructs $g_!Rf'_*K \to Rf_*g'_!K$ as

$$g_!f'_*\mathcal{J}^\bullet = g_!f'_!\mathcal{J}^\bullet = f_!g'_!\mathcal{J}^\bullet = f_*g'_!\mathcal{J}^\bullet \to f_*\mathcal{I}^\bullet$$

Choose a quasi-isomorphism $(a')^{-1}\mathcal{J}^\bullet \to \mathcal{J}_1^\bullet$ to a K-injective complex $\mathcal{J}_1^\bullet$ of sheaves of Λ-modules on $X'_{1,étale}$. Then we can pick a diagram of complexes

$$\begin{array}{ccccc} g'_{1,!}\mathcal{J}_1^\bullet & & \longrightarrow & & \mathcal{I}_1^\bullet \\ \uparrow & & & & \uparrow \\ g'_{1,!}(a')^{-1}\mathcal{J}^\bullet & = & a^{-1}g'_!\mathcal{J}^\bullet & \longrightarrow & a^{-1}\mathcal{I}^\bullet \end{array}$$

commuting up to homotopy where all arrows are quasi-isomorphisms, the equality comes from Lemma 0F51, and $\mathcal{I}_1^\bullet$ is a K-injective complex of sheaves of Λ-modules on $X_{1,étale}$. The map $g_{1,!}Rf'_{1,*}(a')^{-1}K \to Rf_{1,*}g'_{1,!}(a')^{-1}K$ is given by

$$g_{1,!}f'_{1,*}\mathcal{J}_1^\bullet = g_{1,!}f'_{1,!}\mathcal{J}_1^\bullet = f_{1,!}g'_{1,!}\mathcal{J}_1^\bullet = f_{1,*}g'_{1,!}\mathcal{J}_1^\bullet \to f_{1,*}\mathcal{I}_1^\bullet$$

The identifications across the 3 equal signs in both arrows are compatible with pullback maps, i.e., the diagram

$$\begin{array}{ccccc} b^{-1}g_!f'_*\mathcal{J}^\bullet & \longrightarrow & g_{1,!}(b')^{-1}f'_*\mathcal{J}^\bullet & \longrightarrow & g_{1,!}f'_{1,*}(a')^{-1}\mathcal{J}^\bullet \\ \| & & & & \| \\ b^{-1}f_*g'_!\mathcal{J}^\bullet & \longrightarrow & f_{1,*}a^{-1}g'_!\mathcal{J}^\bullet & \longrightarrow & f_{1,*}g'_{1,!}(a')^{-1}\mathcal{J}^\bullet \end{array}$$

of complexes of abelian sheaves commutes. To show this it is enough to show the diagram commutes with $g_!, g_{1,!}, g'_!, g'_{1,!}$ replaced by $g_*, g_{1,*}, g'_*, g'_{1,*}$ (because the shriek functors are defined as subfunctors of the $*$ functors and the base change maps are defined in a manner compatible with this, see proof of Lemma 0F55). For this new diagram the commutativity follows from the compatibility of pullback maps with horizontal and vertical stacking of diagrams, see Sites, Remarks 0F6X and 0F6Y so that going around the diagram in either direction is the pullback map for the base change of $f \circ g' = g \circ f'$ by b. Since of course

$$\begin{array}{ccc} g_{1,!}f'_{1,*}(a')^{-1}\mathcal{J}^\bullet & \longrightarrow & g_{1,!}f'_{1,*}\mathcal{J}_1^\bullet \\ \| & & \| \\ f_{1,*}g'_{1,!}(a')^{-1}\mathcal{J}^\bullet & \longrightarrow & f_{1,*}g'_{1,!}\mathcal{J}_1^\bullet \end{array}$$

commutes, to finish the proof it suffices to show that

$$\begin{array}{ccccccc} b^{-1}f_*g'_!\mathcal{J}^\bullet & \longrightarrow & f_{1,*}a^{-1}g'_!\mathcal{J}^\bullet & \longrightarrow & f_{1,*}g'_{1,!}(a')^{-1}\mathcal{J}^\bullet & \longrightarrow & f_{1,*}g'_{1,!}\mathcal{J}_1^\bullet \\ \downarrow & & \downarrow & & & & \downarrow \\ b^{-1}f_*\mathcal{I}^\bullet & \longrightarrow & f_{1,*}a^{-1}\mathcal{I}^\bullet & & \longrightarrow & & f_{1,*}\mathcal{I}_1^\bullet \end{array}$$

commutes in the derived category, which holds by our choice of maps earlier. $\square$